# Models that remember too much



## AI security · Level 3 · Lesson 2

# Today we will

- Explain memorisation and membership inference.
- Remove identifiers from text before it is used.
- Weigh privacy against usefulness, and ask the right questions.

# Remember

- Rare details can be memorised and repeated word for word.
- Once data trains a model, it is hard to take out.
- Keep it out; redact what must go in; test with a canary.

# Black it out

1. Warm-up (5 min): What could a register tell a stranger?
2. Teach (8 min): How models memorise; what counts as an identifier.
3. Black it out (12 min): Groups remove every identifier.
4. Swap and check (8 min): Find what the other group missed.
5. Wrap-up (7 min): Agree a rule, then the exit ticket.

**Use only made-up names and numbers, never a real register. Hide two or three rare details (“the only pupil who walks from the river village”) so groups learn that identifiers are not only names and numbers.**

# Let’s discuss

1. Which identifiers were hardest to spot, and why?
2. Why can't we just delete the file after a model has learned from it?
3. When is it worth making a model less useful to protect privacy?

# Exit ticket

- Write three kinds of identifier, and one question to ask before any pupils' data goes to an AI service.